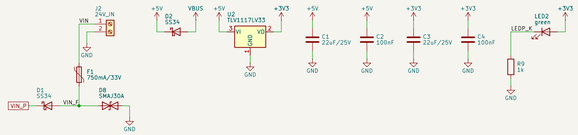
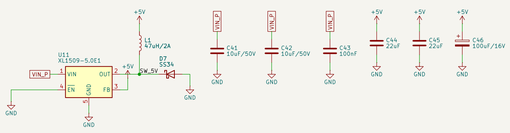
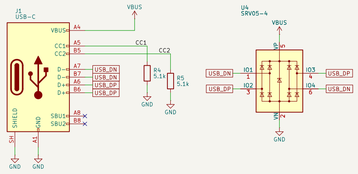
POWER IN 24 VDC terminal or USB
F1+D8 clamp-and-blow at the entry; D1 blocks reverse polarity.
U11 (right) makes +5V; D2 ORs in USB VBUS; U2 drops it to +3V3



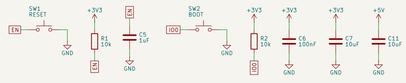
24 V -> 5 V BUCK XL1509-5.0, fixed output
C41–C43 in, D7 catch, L1 47uH, C44/C45/C46 out. FB ties to the output because the part is the fixed 5.0 V version; EN is active LOW and grounded. C46 is an ELECTROLYTIC on purpose: voltage-mode with fixed compensation needs its ESR zero, and an all-ceramic output has none. Keep C41/C42 – U11 – D7 tight: that loop, not L1, is what radiates.



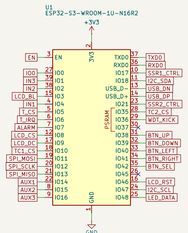
USB-C
native USB flashing + ESD



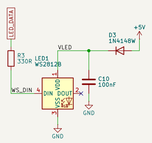
RESET / BOOT / DECOUPLING



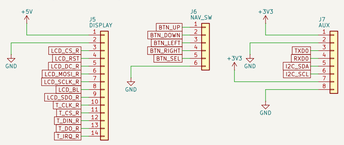
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



WS2812B STATUS LED
VDD dropped ~4.6V for 3.3V data margin

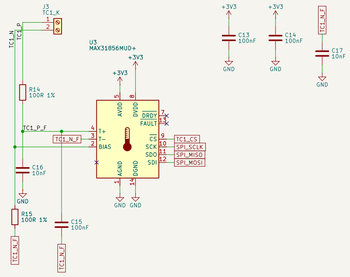


HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C

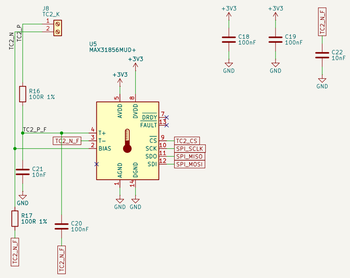


ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults

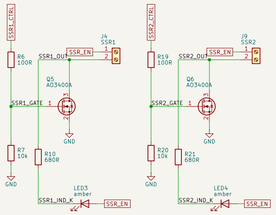
THERMOCOUPLE 1 (control) MAX31856
T– floats to J3, biased via BIAS



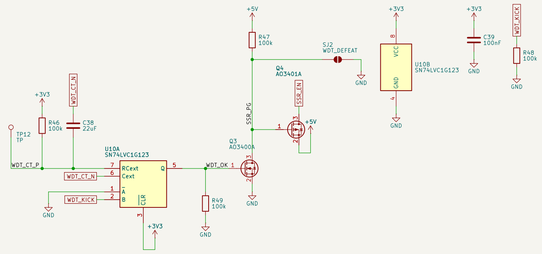
THERMOCOUPLE 2 (load) MAX31856
T– floats to J8, biased via BIAS



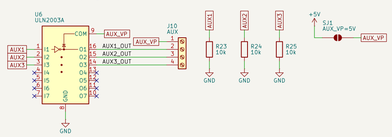
SSR DRIVE x2
low-side AO3400A; indicator LED across each terminal



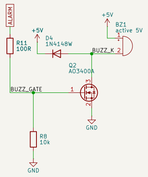
HARDWARE WATCHDOG
U10 retriggerable one-shot: every rising edge on WDT_KICK restarts a ~2.2 s window, so Q stays high only while firmware keeps kicking. Q holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN – the +5V rail feeding BOTH SSR terminals and both indicators.
R46/C38 set the window. R48 holds B low when the MCU is high-Z.
R47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.



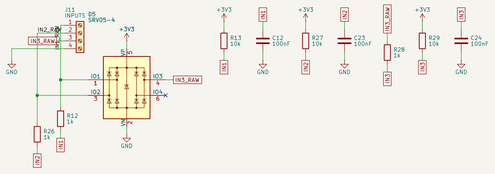
AUX OUTPUT BANK
ULN2003 vent/purge/spare; COM->AUX_VP, SJ1 links to +5V



ALARM BUZZER



PROTECTED INPUTS x3 lid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS



NOTES – Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated). pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NOT opto-isolated – rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K– – both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required; two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

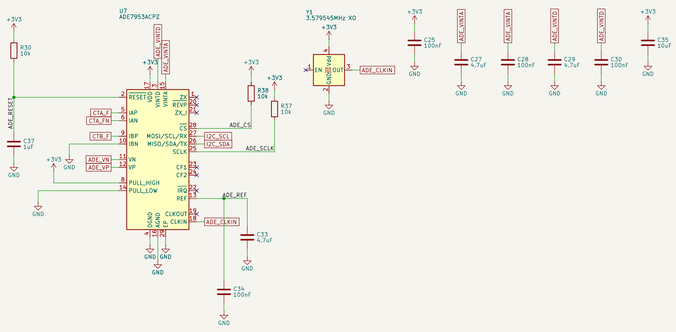
J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND – dry contact each channel to GND.

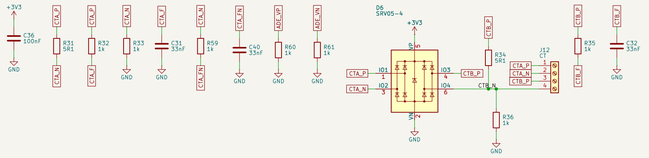
Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V, NOT 3V3 – the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39–R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N – one CT clamp per SSR zone.

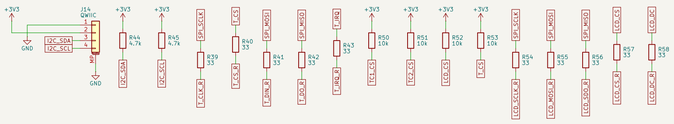
J13: DNP SELV voltage sense header, NOT mains-rated, not fitted – do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED value (unverified C_L); see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.

CT CURRENT SENSING ADE7953, I2C, current-only
no mains – VP/VN to DNP J13





SPI BUS DAMPING + CHIP-SELECT PULL-UPS + I2C EXPANSION
R39–R43 damp the display module's XPT2046 touch lines and R54–R58 the four 40 MHz display lines; both loom to J5. R56 is also the SDO hedge – lift it if the panel does not tri-state J5.9 off the shared SPI_MISO.
R50–R53 hold all four chip selects deasserted from power-on, before firmware configures any GPIO.
J14 Qwiic and J7 5–8 (0.1 in) share I2C, pulled up by R44/R45



TEST POINTS



MOUNTING / FIDUCIALS / POWER FLAGS
FID1–3 are bare-copper pick-and-place targets, on no net



4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project
Sheet: /
File: bisque-controller.kicad_sch
Title: Bisque Kiln Controller
Size: A1 | Date: 2026-09-19 | **Rev: B**
KiCad E.D.A. 10.0.6 | Id: 1/1